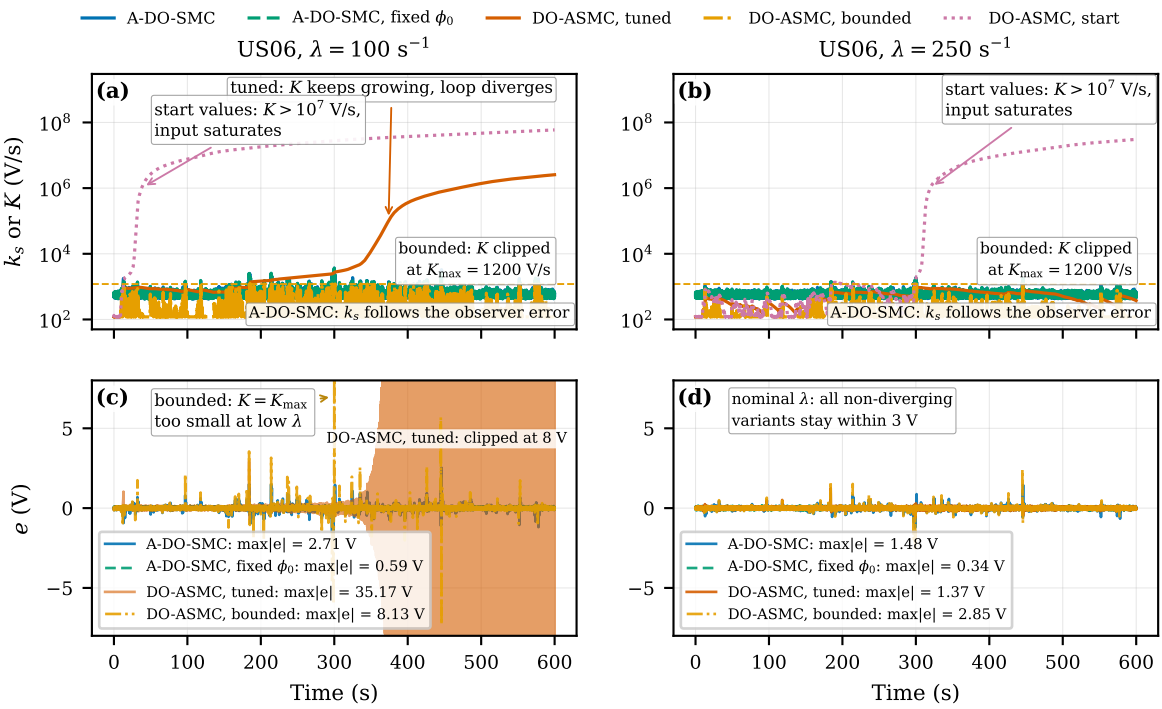

A-DO-SMC
A-DO-SMC, fixed ϕ_0
DO-ASMC, tuned
DO-ASMC, bounded
DO-ASMC, start
US06, $\lambda = 100$ s^{-1}
US06, $\lambda = 250$ s^{-1}
(a)
(b)
(c)
(d)
tuned: K keeps growing, loop diverges
start values: $K > 10^7$ V/s, input saturates
bounded: K clipped at $K_{max} = 1200$ V/s
A-DO-SMC: k_s follows the observer error
k_s or K (V/s)
bounded: $K = K_{max}$ too small at low λ
DO-ASMC, tuned: clipped at 8 V
nominal λ: all non-diverging variants stay within 3 V
e (V)
A-DO-SMC: max|e| = 2.71 V
A-DO-SMC, fixed ϕ_0: max|e| = 0.59 V
DO-ASMC, tuned: max|e| = 35.17 V
DO-ASMC, bounded: max|e| = 8.13 V
A-DO-SMC: max|e| = 1.48 V
A-DO-SMC, fixed ϕ_0: max|e| = 0.34 V
DO-ASMC, tuned: max|e| = 1.37 V
DO-ASMC, bounded: max|e| = 2.85 V
Time (s)